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Section A
Data Fundamentals

1a) What is a database (DB)

It is a digital repository which is used for storing, managing and securing organised collection of data. and manages real time data

They contain any type of data

They store data and organise the data so that it becomes easier to retrieve, manage and analyse the data.

b) Why are databases essential in the modern world?

Every single business or organisation collects data and the data needs to be store, organised, managed and analysed for decision making purposes.

c) Examples of databases

PostgreSQL which is an open-source relational database system used to store and manage structured data.

Oracle Autonomous Database is a cloud database that is designed to run in the cloud.

\# Banks and retail shops make use of databases.

2) What is a data Warehouse (DW)

\# A central repository that:
- collects data
- cleans the data
- stores the data

\# The data is retrieved from multiple sources to support all reporting and analytical data. The data warehouse is built for complex querying and analysis

What is the difference between a database and data warehouse?

Database	Data Warehouse
# Collects data a information	#Data is stored for analytical processing
# It stores current data	# Stores current and historical data from one or more system
# It contains operational and transactional data.	# The data stored may not be up to date as it is based on the frequency of the E.TL
# It contains real time data.	# They use online ~~tra~~ analytical processing
# They use online trans-actional processing	# Stores smaller data quantities that are accessed when needed.
# Stores large amounts of information that must be accessible any time	

Why would a company use a data warehouse

The users of the data warehouse, would use the data to run reports, analyse the data and make decisions based on the data.

3) What is a data lake?

A centralized repository that stores vasts amount of raw data in it's original format.

It stores structured and unstructured data @ any scale.

Data ~~Warehouse~~ Lake	Data ~~Lake~~ Warehouse
# Captures relational and non-relational data from various sources.	# It Captures relational data.
# It contains structured, semi-structured and unstructured data.	# It contains Structured data.
# It contains raw, unfiltred data	# It contains process and vetted data.

Businesses use a data lake for the following reasons:

↳ They serve as a comprehensive platform for storing and processing structured and unstructured data.

↳ It accomodates the storage of raw, unstructured data as it helps businesses manage diverse data types and always the adaptation of ~~exiting~~ evolving data requirements.

(A)

↳ They are designed to scale horizontally, which enables organisations to handle growing volumes of data.

4) <u>What is a Data Mart?</u>

A subset of a data warehouse which focuses on a particular business, department or subject area.

They make specific data available to a defined group of users

It is built from an existing data warehouse It ~~ate~~ allows a business line to discover more focused insights quicker, as opposed to having to work with a broader data warehouse

Uses of a data mart

marketing team to analyze consumer behaviour

Sales team to analyze company / product sales

5) <u>Structured vs Semi Structured vs Unstructured Data</u>

Structured Data	Unstructured Data	Semi-Structured Data
# highly organized quantitative information that fits into fixed rows and columns with a predefined schema	# Qualitative, raw or unorganized information	# It sits between structured data and unstructured data formats # It includes markers or metadata that defines relationships.

Examples of structured ; semi structured data + unstructured data.

1. Structured Data : SQL databases ; Excel ~~da~~ Spreadsheets.

2. Semi Structured : JSON ; HTML

3. Unstructured : Emails ; Videos.

6) What is Data Modeling

It is the process of designing how data will be stored and connected in a system.

It improves data organization and system performance

There is conceptual model and logical model

7) What is Data mining

It is the process of finding ~~problems~~ patterns and useful insights from large datasets

Examples

1. Retail stores analyzing customer buying habits

2. Banks detecting Fraudulent transactions

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8) What is a DBMS?

It is a software used to create and manage databases.

↳ MySQL
↳ Oracle

9) What is ETL?

Extract - Collect data from sources

Transform - Clean and organize the data.

Load - Store the data into a system

ETL helps to transform and prepare the data for analysis

10) Common Data Storage Formats

CSV - Simple text-based table Format
JSON - Lightweight format used in APIS

Section A

References

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Section B

Articial Intelligence and Machine learning

1) What is Articial Intelligence

#It is a branch of computer science focused on creating smart machines that can perform tasks that typically require human intelligence

#AI learns from vast amounts of data, identifying patterns to make predictions or decisions without being explicitly programmed for every scenario

Narrow AI	General AI
# Weak AI # This is the only type of AI that exists today. # It can be trained to perform a single or Narrow task # It targets a single subset of cognitive abilities and advances in the spectrum # Siri	# Strong AI # It is a theoretical concept # It is the next step in the future of AI # The Aim of General AI is to replicate human-level intelligence and resoning # Chatbots.

②

2) What is Machine Learning

It is a branch of AI that focuses on developing models and algorithms that let computers learn from Data without being explicitly programmed for every task.
Teaching the systems to think and understand like humans by learning ~~for~~ from Data

① Supervised learning: Trains models on labeled data to predict or classify new, unseen data.

② Unsupervised learning: finds patterns or groups in unlabeled data.

③ Reinforcement learning: Learns through trial and error to maximize rewards that are ideal for decision making.

3) What is deep learning

It is a subset of machine learning driven by multilayered ~~neutral~~ neutral networks whose design is inspired by the structure of the Human Brain
It powers most of the state of the art AI used today

What are Neutral works

It is a machine learning model that stacks simple "neurons" in layers and learns pattern recognizing weights and biases from Data to map inputs to outputs.

4) What is Natural Language Processing?

\# It is a subfield of computer science and AI that uses machine learning to enable computers to understand and communicate with human Language.

\# It makes it easier for humans to communicate and collaborate with machines by allowing them to do so in the natural Human Language they use daily.

- ↳ Spam detection
- ↳ predictive text and autocorrect.
- ↳ google translate.

5) What is Large Language Model

It is an AI model trained on huge amounts of text data

- → CHAT GPT
- → Claude

6) What is Generative AI

It creates new content like text, images, music or videos.

7) What is a training Dataset

#It is the data that is used to teach an AI model.

#Good quality data is important because bias data can produce inaccurate results.

8) What is Computer Vision?

It allows a machine to understand images and videos.

Section B

References

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Section C

1) What is an API

Application programming interface
It is a set of rules that allows software applications to communicate to each other to exchange data; features and functionality.
They enable developers to integrate data, services and capabilities from one application to another
Data and functions can be made available to internal and external users.
For Example customers can buy DStv via MWeB. Instead of going to DStv to get a DStv subscription they can go to mweb to buy a DStv data bundle. API's are used between Mweb and DStv to create the customer and provide them with DStv.

2) What is an API key?

It is a unique identifier used to authenticate software and systems attempting to access other software or systems.
APIs provide users with sensitive data and it is important to ensure that the API can validate the application making the request to access the information is allowed access the information.
It ensures that an organizations data remains secure.

multiple authentication methods can be used to secure API keys such as:
→ Secure key storage
→ rotating and generating new API keys every XX number of days help to keep the APIs secure.

if the API key is not secured, it can enable unauthorized access to the company's API which can lead to unauthorized usage, data exposure, service abuse, reputational damage.

3) What is a R.E.S.T API

Representational State Transfer
It is a set of web API architecture principles
They use HTTP requests such as G.E.T, PUT, HEAD and Delete to interact with resources
They communicate through HTTP requests to perform standard database functions like
→ creating records
→ reading records
→ updating records
→ deleting records

1. G.ET request would be used to retrieve information
2. P.O.ST request would create a new record
3. P.U.T request would update a record
4. Delete request would a record.

4) What is JSON?

#Javascript Object Notation

It is mostly used for data storage and transfer

It's used by web developers for transferring data between a server and a web application

Its format makes it ideal for exchanging data across different programming languages and platforms.

It is a lightweight format for storing and transferring data.

Easy to understand

~~{ " employees" : [{" firstNam~~

```
{*
 " employees : [
 { " First Name " : " John ", "LastName : "Doe " },
  }
  }
```

5) What is prompt engineering?

It is the skill of writing effective instructions for AI tools

6) What is a MAP function?

It applies the same operation to every item in a list.

7) What is cloud Computing?

It is providing computing services over the internet.

Saas : Software as a Service

Paas : Platform as a Service

8) What is Version Control?

It tracks changes in code and Files.

9) What is a Jupyter Notebook?

An interactive enviroment used for coding and analysis.

10) What is Python?

A programming Language used in data analytics

Section C

References

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Section D

Data Ethics, Governance and Careers

1) What is Data Privacy? Why is it important?

It is the proper handling, processing, storage and usage of personal information.

Companies regularly collect a users data like email addresses, phone numbers, physical addresses

Data privacy empowers users to control how their data is collected, processed, stored and shared.

Data privacy protects the users personal information against identify theft, financial fraud and unauthorised transactions.

What is the GDPR and which countries use it?

It is the General Data Protection Regulation

It is a comprehensive European Union law that specifies how data should be collected, processed stored and used.

It specifies the rights of the consumer and the rights of the business.

Italy, France, Germany

2) What is Data Ethics

Ethics that evaluate data practices

It guides how an organisation collects, shares, and uses data especially personal data.

It aims to make sure that data is used fairly, in a non-discrimatory way

① Google # Google provided the users location data to advertisers.
This indicates that data was missused and there were privacy breaches.

② Twitter # Twitter provided advertisers with their users personal data without their consent
This was violation of the users privacy

③ Facebook and Cambridge Analytica : Data from Facebook users was obtained for political profiling and targeting without obtaining consent.

3) What is Data Governance

It is the data management principle that focuses on the quality, security and availability of a company's data.

The components include:

① Data owners : This is the data domains

② Data stewards : This is the handling of ~~ad~~ daily management of specific data domains

③ Stakeholders : The people who use the data.

A) What is Bias in AI

This is machine learning bias or algorithm bias
It is due to the occurrence of biased results due to human biases that skew the original training data.
If these biases are not addressed, they impact the organisation's success and their ability to participate in the economy and society
It reduces AI's accuracy.

Example : Healthcare
→ Computer-aided diagnosis systems have been found to return lower accuracy results for African American patients than white patients.
→ The results are inclusive of all races and mis-represented groups.

5) Data Engineer	Data Analyst	Data Scientist	AI Engineer
Builds and maintains the data infra-structure.	Looks @ the performance of a business by looking at the past and current Data	Predicts the future of a business by using data.	Builds AI powered applications and automation
Creates data pipelines, moves data between systems	Cleans the data; creates reports & dash-boards, looks for trends.	Builds predictive models	Integrates AI models into products, develops chatbots, & AI apps

Data Engineer	Data Analyst	Data Scientist	AI Engineer
ETL, data-bases	Excel, SQL	Phython / R	Open AI, API's, PyTorch

6) What is Business Intelligence

The process of collecting, analyzing and visualizing data to help businesses make better faster and more informed decisions
↳ Power BI
↳ Qlikview

It helps businesses analyze their data.

Section D
Refers

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6) Ibm.com/think/topics/data-governance
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Section E

1) I Found the API section interesting as I work with APIS.

2) I Found the data Governance section challenging. Because I ~~t~~ thought I knew it but I actually do not know it that well.
↳ I used the GDPR website.

3) * Data will help me make informed decisions.
\# It will help me turn my numbers into meaning insights.
\# I work with data at work it will help me identify patterns and trends.
* AI is a tool that will help me work faster and also challenge my work.